Automatic Detection and Correction of Defective Pixels in PRISMA Hyperspectral Data

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***Abstract*— In satellite hyperspectral sensors, a standard procedure based on homogeneous reference sources is used to regularly update the map of defective pixels (DPs). Unfortunately, this procedure often fails to detect some DPs both because they may arise in the interval between two standard calibration steps and because they may be characterized by subtle or unexpected signal values. The resulting hyperspectral image is affected by residual nonuniformity noise, which correlates in the along-track direction. This noise source reduces the quality of hyperspectral products, such as classification, unmixing, and material detection. In this article, we present a new procedure to find the location of the residual DPs in the detection matrix and we also propose an effective method to estimate the missing radiance values inferring them from the image pixels by leveraging both spatial and spectral correlation. The procedure, here tailored to PRecursore IperSpettrale della Missione Applicativa (PRISMA) hyperspectral images, is quite general and can be easily adapted to process images recorded by any satellite pushbroom hyperspectral sensor. The improved image quality yielded by the proposed procedure is first demonstrated qualitatively, by comparing the global RX (GRX) maps on the original image with those obtained after detection of the DPs and correction of their radiance values. The image quality improvement is then quantified on a set of nine PRISMA images, recorded in an interval of about two years, using two *ad hoc* defined indexes. The analysis is carried out separately for the visible and near-infrared (VNIR) and shortwave infrared (SWIR) spectrometers of the PRISMA mission.**

***Index Terms*— Defective pixel (DP) detection, hyperspectral image, PRecursore IperSpettrale della Missione Applicativa (PRISMA), residual nonuniformity.**

I. Introduction

PUSHBROOM hyperspectral sensors undergo careful ground laboratory measurements in order to characterize their radiometric properties and the spectral–spatial mapping. However, due to aging or degradation of optical or electronic components and mechanical vibrations, instrument characteristics may change and the instrument has to be recalibrated at regular intervals [1], [2]. This regular recalibration process is more effective on airborne sensors, which can be easily accessed with accurate instrumentation before the flight, but it is more complicated in spaceborne sensors. In spaceborne sensors, nonuniformity noise is compensated through the flat field correction process that exploits calibration sources derived from natural uniform targets (e.g., Sun, Moon, desert, and ice covered areas) [3]. Flat field correction allows also the detection of defective detectors, i.e., detectors that have an anomalous or abnormal response [4]. Such information is then used to build a defective pixel mask (DPM) where the position of defective detectors in the sensor acquisition matrix is annotated. In Level 1 (L1) data, the values of the defective pixels (DPs) are corrected through interpolation. Unfortunately, the DPM derived in orbit is often not accurate enough to allow the detection of all the DPs and, even worse, the number and position of the DPs may change with time [5], i.e., the DPM can change from an image to another. Missed DPs detection results in the emergence of a noise source in the image, called residual fixed pattern noise (RFPN), which may cause evident stripes in the along-track (ALT) direction in the spectral channels containing DPs. However, DPs experienced in real hyperspectral images also include artifacts that do not necessarily show up as continuous and easily detectable stripes. RFPN is perhaps the most serious noise source and can impair the performance of algorithms based on the integration of multiple spectral channels, such as those adopted for anomaly detection, classification, object detection, and abundance map estimation [1], [6], [7].

The difficulties encountered in properly updating the DPM and correcting the missing data corresponding to DPs are well known and have been experienced in previous hyperspectral satellite missions such as Nasa's EO1 Hyperion [4] and ESA's Chris/Proba [8]. In this work, we focus on the hyperspectral sensor of the Earth observation mission of the Italian Space Agency (ASI) named PRecursore IperSpettrale della Missione Applicativa (PRISMA) [9], [10], [11], [12]. The mission has been operational since March 2019 and data for the community are available according to the ASI data policy [13]. PRISMA payload includes a pushbroom hyperspectral camera operating in the visible and near-infrared (VNIR) and shortwave infrared (SWIR) spectral ranges. Specifically, the HS imaging sensor covers the portion of the electromagnetic spectrum ranging from 400 to 2500 nm with 10-nm spectral sampling, through two partially overlapped spectrometers [12]: one covers the VNIR spectral range (400–970 nm) and has 63 channels and

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the other covers the SWIR spectral range (940–2500 nm) with 171 channels.

As evidenced by the results of post-launch validation [14], PRISMA provides high-quality data. The percentage of DPs detected by the in-flight calibration process is in the order of 0.1% of the total number of image pixels. However, such as other satellite hyperspectral data, PRISMA data suffer from missed detection of DPs. Such DPs show up in terms of unexpected offset or variance with respect to the neighboring pixels that may persist during the whole image acquisition interval or in a short part of it.

To better state the problem and the level of intervention of the procedure proposed in this work, it is worth recalling that L1 PRISMA data are provided at two different levels of processing [2]. The first level, denoted with hyperspectral radiometrically calibrated (HRC), consists of radiance-calibrated data before applying any spatial/spectral resampling. In the second level, denoted as hyperspectral coregistered (HCO), the radiance data are spatially and spectrally resampled in order to correct smile and keystone effects and spatial/spectral misalignments. To generate the HCO product, the pixels in the detector matrix are mixed together by the interpolation process and the residual DPs are combined with their neighbors, thus spreading the nonuniformity and worsening further the data quality. The process of detection and correction of residual DPs is more complicated in the HCO data and it is, therefore, advisable to apply it to HRC data.

To give evidence of the problem, let us consider the PRISMA image whose false-color (FC) representation obtained by combining the channels of red, green, and blue (RGB) is shown in Fig. 1(a). The image refers to the region around the town of Pisa, Italy, and was acquired on June 10, 2021. Since DPs can be considered as spectrally anomalous pixels, they can be highlighted by applying an anomaly detection algorithm such as the well-known global RX (GRX, see [15]). GRX is designed to detect pixels whose spectrum differs significantly from that of the rest of the image pixels and it is expected to be very effective in detecting not only true anomalies but also the subtle anomalies that are generated by a noncorrect response of some detector. For this reason, the GRX map is often used to validate the effectiveness of DPs correction (DPs-C) procedures [7]. Fig. 1(b) shows the output of the GRX applied to the VNIR channels of the HCO image of the considered PRISMA data. The presence of residual DPs is quite evident considering the bright stripes appearing in the ALT direction.

A. Contributions of the Work

In the literature, there is a wealth of papers on the subject of nonuniformity and striping correction (see [16], [17], [18], [19]). In general, such techniques aim at adjusting the values of the gain and offset coefficients for each detector on the acquisition matrix, assuming that they remain unchanged during the acquisition process and by imposing some average homogeneity constraint. In this work, we focus on DPs rather than traditional striping noise, by considering not only those detectors that have anomalous gain and offset values along the acquisition line but also those showing a varying response in the time interval of image acquisition and that cannot be corrected simply by applying constant gain and offset.

Namely, according to the PRISMA processing chain, we consider that the Level 0 product is calibrated by using uniform (natural) sources to adjust the detectors' gain and offset and to identify the most of DPs in the acquisition matrix. These latter are annotated in the known DPM (kDPM) that is supplied by ASI together with the Level 1 data product. Starting from the kDPM, we propose a scene-based and fully automatic algorithm that the following conditions hold.

1) Estimate the missing information for the DPs included in kDPM.

2) Refine the DPM by identifying new DPs that showed up during the image acquisition or for some reason are not detected by the in-flight flat field calibration process.

3) Use the updated DPM to drive the estimation of the missing radiance values in the newly detected DPs.

Thus, different from most striping correction approaches (see [16], [17], [19], [20]), here the estimated radiance values are used only to replace the missing values in a small percentage of pixels (those corresponding to the DPs), while the other values are left unchanged.

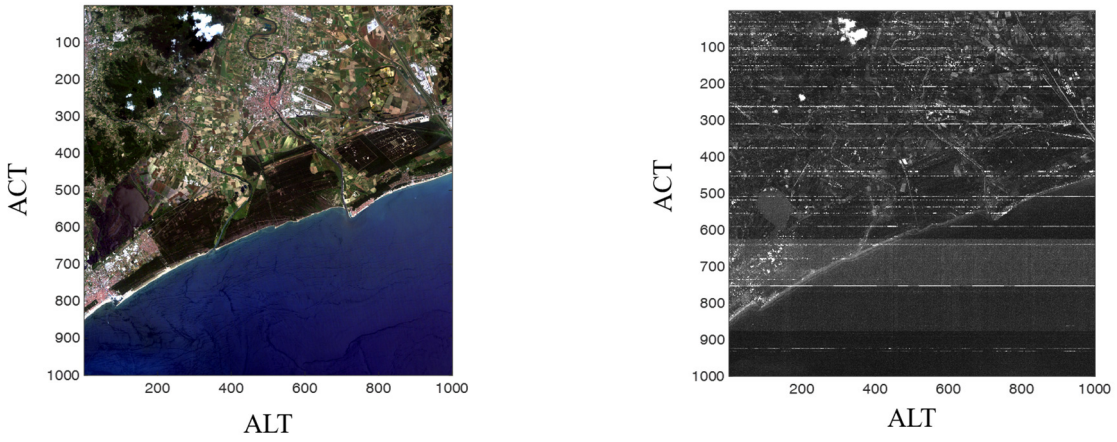
Only a few works proposed in the literature share this general approach. In [21], DPs in Hydice data are detected based on the signal-to-noise ratios and the linearity of the response. DPs replacement is performed using neighboring pixels (both spectral and spatial) not marked as defective. In [5], a method to identify DPs was proposed based on the assumption that neighboring cross-track pixels respond similar to one another along the flight line. The method is only applied to the correction of the offset term starting from dark frames measurements. Both methods in [5] and [21] exploit illuminated and/or dark (shuttered) measurements and cannot be considered as scene-based methods. Fischer *et al.* [7] assumed constancy of the spectral ratios to identify DPs in SWIR hyperspectral data and to correct them. The method is claimed to be computationally effective and robust to image noise. However, the authors themselves admit that the performance of the method based on band ratios might be limited in highly cluttered areas causing errors in DPs location and reducing the accuracy in the estimate of the missing values.

The strategy proposed in this work differs from existing solutions (see [5], [7], [21]) both for the method used to estimate the missing radiance values and for the strategy to detect the location of DPs not reported in the kDPM.

For a given DP, missing radiance values are estimated by leveraging both the spatial and the spectral correlation of hyperspectral data and so as not to add further artifacts or delete important features related to rare anomalous objects in the scene.

Detection of DPs not present in the kDPM exploits the ability of the GRX to highlight anomalous pixels and, in particular, exploits the ALT persistence of the anomaly generated by a defective sample to identify it and to start the search

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(a) (b)

Fig. 1. (a) RGB representation of the PRISMA image acquired on June 10, 2021 and (b) output of the GRX algorithm applied to the VNIR HCO image. In both the figures, the y-axis is the ACT direction and the x-axis is the ALT direction.

of the band coordinate of the DP. The proposed method has proved effective in detecting DPs characterized by anomalous and constant gain and offset values, those whose gain and offset vary during the acquisition time and also DPs affected by such subtle disturbances to be not easily appreciable by visual inspection of the single-band image.

Importantly, the proposed procedure applies to L1 HRC PRISMA data in order to better detect DPs and avoid artifacts caused by the spatial and spectral resampling required to get the L1 HCO data.

The rest of this article is organized as follows. In Section II, we introduce the mathematical model for the hyperspectral image pixel, including the RFPN due to DPs. Section III describes the algorithm proposed to restore the image values in the DPs recorded in the kDPM, while Section IV is dedicated to the description of the algorithm to detect and correct the DPs not reported in the kDPM. In Section V, the results obtained on real PRISMA images are presented and discussed. Finally, in Section VI, we draw the main conclusions of our work and map the road for future developments.

II. Hyperspectral Image Model

In this section, we introduce the model adopted for the hyperspectral image pixels, including the RFPN due to DPs.

Throughout this article, the two dimensions of the sensor acquisition matrix are named band and sample. The term band refers to the spectral dimension, while sample refers to the spatial across-track (ACT) dimension. We denote as B and S the total number of bands and samples, respectively. The temporal dimension (ALT direction) is referred to as a line and the total number of lines is denoted as L. In the case of PRISMA data, both spectrometers have $S = 1000$ and $L = 1000$. The number of bands is $B = 63$ for the VNIR spectrometer and $B = 171$ for the SWIR one.

The ACT and the spectral coordinates of a given DP are indicated as defective sample and defective band, respectively.

In general, we use bold lower case to denote vectors, bold upper case to denote matrices, and italic font for scalar variables. The notation $\mathbf{w}(s,l) \in \mathcal{R}^{Bx1}$ denotes the $Bx1$ spectral pixel in the sample s and line l of a given image, while $\mathrm{w}_b(s,l)$ denotes the value of the same pixel in the bth band. $\mathbf{W}(s) \in \mathcal{R}^{BxL}$ is the matrix whose columns are all the spectral pixels in the line addressed by s, i.e., $\mathbf{W}(s) = [\mathbf{w}(s,1), \ldots, \mathbf{w}(s,L)]$.

Denoting as $\mathbf{y}(s,l)$ the spectral pixel of the HRC PRISMA image, we assume that

$$\mathbf{y}(s,l) = \mathbf{x}(s,l) + \mathbf{n}(s,l) + \mathbf{d}(s,l) \tag{1}$$

where $\mathbf{x}(s,l)$ is the useful signal corresponding to the at-sensor spectral radiance, $\mathbf{n}(s,l)$ is the random noise modeled as a Gaussian random vector having zero mean and diagonal covariance matrix $\mathbf{\Lambda} = \boldsymbol{diag}(\sigma_1^2, \ldots, \sigma_B^2)$, and $\mathbf{d}(s,l)$ is the RFPN noise vector modeling the DPs. DPs are reported in the DPM denoted as $\mathbf{\Psi} \in \mathcal{R}^{BxS}$. $\mathbf{\Psi}$ is a binary matrix whose element $\Psi(b,s)$ is equal to 1 if the detector at (b,s) is defective and 0 otherwise. Since the percentage of defective detectors is generally very low (<1% of the total), $\mathbf{\Psi}$ is a sparse matrix. Consequently, $\mathbf{d}(s,l)$ is also sparse. Specifically,

$$d_b(s,l) = \begin{cases} 0, & \text{if } \Psi(b,s) = 0 \\ G(l)\cdot x_b(s,l) + O(l), & \text{if } \Psi(b,s) = 1. \end{cases} \tag{2}$$

Note that the residual gain $G(l)$ and the offset $O(l)$ are not necessarily the same for each defective detector, so they should depend on b and s. However, for the sake of conciseness, in (2) and in the following, we omit this dependence.

Based on the characteristics of $G(l)$ and $O(l)$, DPs can be roughly grouped into three main classes.

1) *Cold DPs* and *Hot DPs*: They occur when $G(l) \cong -1$ and $O(l)$ is much lower (cold DP*s*) than the random noise standard deviation or much higher (hot DPs) than the mean value of the useful signal. They are originated by defective detectors having very low or high sensitivity and in general are easily detected by the in-flight calibration process.

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2) *Linear DPs*: They occur when both $G(l)$ and $O(l)$ remain constant during the acquisition time ($\forall l = 1, \ldots, L$) and are generally originated by detectors with anomalous and constant (over the acquisition time) response. Usually, this class of DPs can be easily detected when their responses significantly differ from those of the neighboring detectors.
3) *Irregular DPs*: They occur when $G(l)$ and $O(l)$ vary with the acquisition time l. Such DPs are generated by unstable detectors whose response undergoes variation during the acquisition scan. In some cases, their malfunction can be limited to a subinterval of the entire acquisition time. In general, these DPs are harder to detect due to the unpredictable behavior of the gain and offset values.

The PRISMA in-flight calibration process is able to detect most of the DPs belonging to the first two classes. In this process, the radiance value in each detected DP is set to zero and the DPs position is annotated in the kDPM, denoted as $\bar{\boldsymbol{\Psi}}$ hereinafter.

Regardless of the class they belong to, image pixels corresponding to the DPs have an anomalous spectral behavior. Thus, they can be highlighted by using a (spectral) anomaly detection algorithm such as the GRX whose output in each position (s, l) is obtained as

$$\mathrm{GRX}(s,l) = \left\| \boldsymbol{\Gamma}^{-\frac{1}{2}} \left[\boldsymbol{y}(s,l) - \boldsymbol{\mu} \right] \right\|_2^2. \tag{3}$$

In (3), $||\,||_2$ is the Euclidean norm ($L2$-norm), $\boldsymbol{\mu} \in \mathcal{R}^{Bx1}$ and $\boldsymbol{\Gamma} \in \mathcal{R}^{BxB}$ are the mean spectral vector and the covariance matrix estimated using all the image pixels, respectively. It is worth noting that, since DPs are a very small fraction of the entire image pixels, one can reasonably assume that the bias on the estimates of $\boldsymbol{\mu}$ and $\boldsymbol{\Gamma}$ is negligible.

For a given ACT coordinate (s) corresponding to a DP, $\mathrm{GRX}(s, l)$ is expected to have high values for all the ALT coordinates (l) affected by the anomalous behavior of $G(l)$ and $O(l)$. This makes GRX a useful tool to evaluate the effectiveness of a given algorithm to estimate the missing values in the DPs [7]. In addition, GRX can be used to guide the detection of those DPs that are hardly detectable by the in-flight calibration process.

The procedure proposed in this work has two main functional blocks. The first, called DPs-C, starts from $\bar{\boldsymbol{\Psi}}$ and estimates the missing radiance values in the DPs. The second block, called DPs detection and correction (DPs-DC), exploits the information provided by the GRX output to detect the DPs not reported in $\bar{\boldsymbol{\Psi}}$ and restores the corresponding missing radiance values. In the following, we describe the two steps of the proposed procedure. To better explain the various elements of the processing chain, we make use of examples referring to the PRISMA hyperspectral image introduced in Section I.

III. DPs Correction

Since the useful signal in hyperspectral images exhibits a high degree of spectral correlation, we can approximate the vector $\boldsymbol{x}(s, l) \in \mathcal{R}^{Bx1}$ in (1) as lying onto a subspace Φ having dimension T much lower than B [22]. Thus, denoting as $\boldsymbol{\Omega} \in \mathcal{R}^{BxT}$ the rank-T matrix whose columns are the basis vectors for Φ, $\boldsymbol{x}(s, l)$ can be rewritten as

$$\boldsymbol{x}(s,l) = \boldsymbol{\Omega} \cdot \boldsymbol{a}(s,l) \tag{4}$$

where $\boldsymbol{a}(s, l) \in \mathcal{R}^{Tx1}$ is the vector of the representation coefficients on the basis $\boldsymbol{\Omega}$. The proposed DPs-C procedure exploits such a low-rank representation of the radiance data to restore the values in the DPs. To explain the procedure, let us focus on the sample s_{DP} corresponding to one of the DPs in $\bar{\boldsymbol{\Psi}}$ and denote as $\Theta(s_{\mathrm{DP}})$ the set of the c defective bands for the sample s_{DP}, i.e., $\Theta(s_{\mathrm{DP}}) \equiv \{b : \bar{\boldsymbol{\Psi}}(b, s_{\mathrm{DP}}) = 1\}$.

According to the models in (1), (2), and (4), all the pixels in the line addressed by the ACT coordinate s_{DP} can be expressed as

$$\begin{aligned} \boldsymbol{y}(s_{\mathrm{DP}}, l) &= \boldsymbol{\Omega} \cdot \mathbf{a}(s_{\mathrm{DP}}, l) + \boldsymbol{d}(s_{\mathrm{DP}}, l) + \boldsymbol{n}(s_{\mathrm{DP}}, l) \\ \boldsymbol{d}(s_{\mathrm{DP}}, l) &= \sum_{b \in \Theta(s_{\mathrm{DP}})} d_b(s_{\mathrm{DP}}, l) \boldsymbol{e}_b \end{aligned} \tag{5}$$

where $\boldsymbol{e}_b$ is the bth vector of the canonical basis. Denoting as $\mathbf{E}_{\Theta(s_{\mathrm{DP}})} \in \mathcal{R}^{Bxc}$ and $\boldsymbol{d}_{\Theta(s_{\mathrm{DP}})}(s_{\mathrm{DP}}, l) \in \mathcal{R}^{cx1}$, the matrix whose columns are the vectors $\boldsymbol{e}_b$ and the vector whose elements are the values $d_b(s_{\mathrm{DP}}, l)$ for each $b \in \Theta(s_{\mathrm{DP}})$, the DP noise vector $\boldsymbol{d}(s_{\mathrm{DP}}, l)$ can be rewritten as

$$\boldsymbol{d}(s_{\mathrm{DP}}, l) = \mathbf{E}_{\Theta(s_{\mathrm{DP}})} \cdot \boldsymbol{d}_{\Theta(s_{\mathrm{DP}})}(s_{\mathrm{DP}}, l). \tag{6}$$

Accordingly, (5) can be expressed in a more compact form as

$$\begin{aligned} &\boldsymbol{y}(s_{\mathrm{DP}}, l) = \tilde{\Omega} \cdot \boldsymbol{\phi}(s_{\mathrm{DP}}, l) + \boldsymbol{n}(s_{\mathrm{DP}}, l) \\ &\tilde{\Omega}(s_{\mathrm{DP}}) = \left[\boldsymbol{\Omega}, \mathbf{E}_{\Theta(s_{\mathrm{DP}})} \right] \text{and} \boldsymbol{\phi}(s_{\mathrm{DP}}, l) = \begin{bmatrix} \boldsymbol{a}(s_{\mathrm{DP}}, l) \\ \boldsymbol{d}_{\Theta(s_{\mathrm{DP}})}(s_{\mathrm{DP}}, l) \end{bmatrix}. \end{aligned} \tag{7}$$

Since both c and T are typically much smaller than B, we can obtain the maximum likelihood (ML) [23] estimate $\hat{\boldsymbol{\phi}}(s_{\mathrm{DP}}, l)$ of $\boldsymbol{\phi}(s_{\mathrm{DP}}, l)$ from the observation $\boldsymbol{y}(s_{\mathrm{DP}}, l)$ as

$$\hat{\boldsymbol{\phi}}(s_{\mathrm{DP}}, l) = \tilde{\Omega}(s_{\mathrm{DP}})^{+} \cdot \boldsymbol{\Lambda}^{-\frac{1}{2}} \cdot \boldsymbol{y}(s_{\mathrm{DP}}, l) \tag{8}$$

with $\tilde{\Omega}(s_{\mathrm{DP}})^{+} = [\tilde{\Omega}(s_{\mathrm{DP}})^{T} \cdot \boldsymbol{\Lambda}^{-1} \cdot \tilde{\Omega}(s_{\mathrm{DP}})]^{-1} \cdot \tilde{\Omega}(s_{\mathrm{DP}})^{T} \cdot \boldsymbol{\Lambda}^{-(1/2)}$ representing the Moore–Penrose pseudoinverse of $\tilde{\Omega}(s_{\mathrm{DP}})$ in the noise-whitened space.

According to (7),

$$\hat{\boldsymbol{\phi}}(s_{\mathrm{DP}}, l) = \begin{bmatrix} \hat{\boldsymbol{a}}(s_{\mathrm{DP}}, l) \\ \hat{\boldsymbol{d}}_{\Theta(s_{\mathrm{DP}})}(s_{\mathrm{DP}}, l) \end{bmatrix}$$

that suggests restoring the corrupted HRC image pixel $\boldsymbol{y}(s_{\mathrm{DP}}, l)$ as

$$\bar{\boldsymbol{y}}(s_{\mathrm{DP}}, l) = \boldsymbol{y}(s_{\mathrm{DP}}, l) - \mathbf{E}_{\Theta(s_{\mathrm{DP}})} \cdot \hat{\boldsymbol{d}}_{\Theta(s_{\mathrm{DP}})}(s_{\mathrm{DP}}, l) \tag{9}$$

with $\bar{\boldsymbol{y}}(s_{\mathrm{DP}}, l)$ denoting the corrected pixel.

It is worth noting that (9) modifies only the corrupted bands of $\boldsymbol{y}(s_{\mathrm{DP}}, l)$ leaving the others unchanged.

The unknown basis matrix $\boldsymbol{\Omega}$ and the noise covariance matrix $\boldsymbol{\Lambda}$ are estimated by using the well-known HySime algorithm [22] applied to a set of training data extracted

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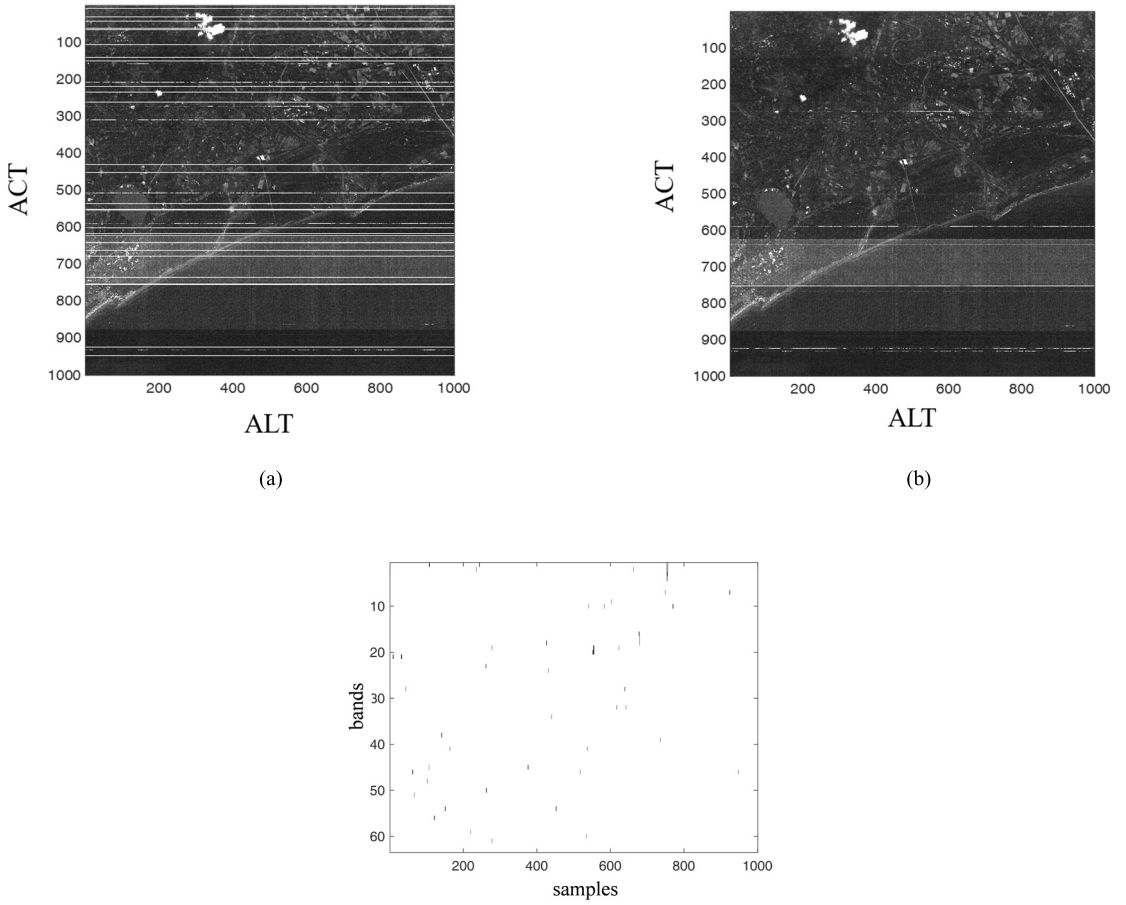


Fig. 2. Output of the GRX obtained on (a) original VNIR HRC image of PISA and (b) image after fixing the pixels that were already known to be defective (DPs-C). (c) kDPM for the VNIR channels, the DPs are marked in black.

from the image by considering all the spectral pixels in a neighborhood Υ of s_{DP} and discarding any DPs, i.e.:

$$\Upsilon \equiv \left\{ s : s \in [s_{\mathrm{DP}} - m, s_{\mathrm{DP}} + m] \quad \text{and} \quad \bar{\mathbf{\Psi}}(b, s) = 0 \ \forall \ b \right\} \tag{10}$$

with m determining the neighborhood size.

Using local training data, instead of the entire hyperspectral image, is expected to yield a better estimate of the lowest rank subspace that fits the one spanned by the useful signals along the lines (l) of the sample s_{DP}.

Of course, the described procedure is iterated for all the defective samples s_{DP} reported in $\bar{\mathbf{\Psi}}$.

To give an example of the effectiveness of the DPs-C procedure and to better introduce the next DPs-DC step, we show the results provided by DPs-C on the HRC PRISMA image presented in Section I. The example refers to the VNIR spectrometer. Fig. 2(a) shows the GRX output of the original HRC product where the DPs generate the evident horizontal bright lines. Fig. 2(b) shows the GRX output obtained after the application of DPs-C. For completeness, in Fig. 2(c), we also show the kDPM $\bar{\mathbf{\Psi}}$ where the DPs are marked in black.

Better product quality obtained by applying DPs-C is evident since most of the bright lines present in Fig. 2(a) fade away moving to Fig. 2(b). However, we can see that some bright lines still remain, which are due to DPs not reported in $\bar{\mathbf{\Psi}}$.

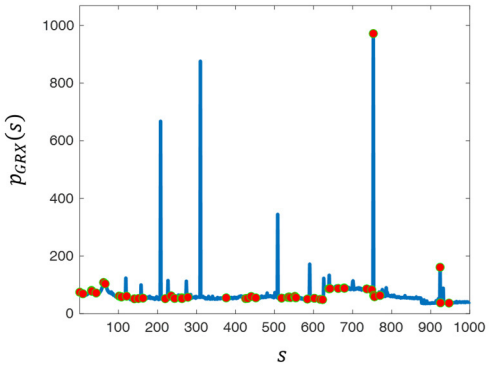


Fig. 3. ACT profile of the GRX $[p_{\mathrm{GRX}}(s)]$ output obtained on the VNIR HRC image after the application of DPs-C. Red plots indicate the defective samples reported in $\bar{\mathbf{\Psi}}$.

In addition, in Fig. 3 with reference to the image provided by DPs-C, we show the ACT profile of the GRX $[p_{\mathrm{GRX}}(s)]$ obtained by averaging its output in the ALT direction. The $p_{\mathrm{GRX}}(s)$ peaks correspond to persistent spectral anomalies in the ALT direction, which are mostly due to DPs. In Fig. 3, we have indicated, with red points, the ACT coordinates (defective samples) of the DPs in $\bar{\mathbf{\Psi}}$. We see that, for almost

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all the known defective samples, we do not have peaks in $p_{\mathrm{GRX}}(s)$. This is an indirect proof of the effectiveness of the DPs-C algorithm. For two of the known defective samples ($s = 753$ and $s = 924$), it might seem that DPs-C does not properly recover the missing values. Actually, such defective samples have an additional defective band not reported in $\bar{\boldsymbol{\Psi}}$. For $s = 753$, $\bar{\boldsymbol{\Psi}}$ reports $b = 1$ and $b = 2$ as defective bands, but we found an additional DP at $b = 24$. Similarly, for $s = 924$, the known defective band is $b = 7$, but we found an additional DP at $b = 9$. Evidence of these observations is given in Fig. 4. Specifically, in Fig. 4(a), we plot the entire line corresponding to $s = 753$ and $b = 24$ of the HRC image together with the two adjacent lines ($s = 752$ and $s = 754$) that are taken as a reference for the uncorrupted signal. Similar plots are shown in Fig. 4(b) with reference to $s = 924$ and $b = 9$. In both cases, the effect of DPs is quite evident.

We would like to note that Fig. 4(a) and (b) provides visual examples of what DPs not recorded in $\bar{\boldsymbol{\Psi}}$ might look like. The specific behavior of those DPs with respect to their neighborhood is not used to detect them. The specific procedure proposed to detect DPs not included in $\bar{\boldsymbol{\Psi}}$ is detailed in Section IV.

We conclude this section with a general remark about the GRX output of the PRISMA images. In Fig. 2, we see a sort of banding effect occurring in the samples from 626 to 750. It is also clearly evident in the GRX output of the HCO data shown in Fig. 1(b) and it has been observed in all PRISMA images analyzed in this article. Banding is quite different from DPs and the two effects are probably generated by different causes. In this work, we focus on the correction of the artifacts caused by DPs and leave the analysis and the mitigation of the banding effect as a topic for future work.

IV. DPs Detection and Correction

To restore the DPs not reported in $\bar{\boldsymbol{\Psi}}$, we propose an automatic algorithm (DPs-DC) based on an iterative procedure. The algorithm detects the defective samples and the defective bands of the unknown DPs, updates the DPM ($\boldsymbol{\Psi}$), and corrects the corrupted radiance values (in the whole ALT direction). The DPM is initialized to $\bar{\boldsymbol{\Psi}}$ ($\boldsymbol{\Psi}^{(0)} = \bar{\boldsymbol{\Psi}}$) and the analyzed data are those obtained after the DPs-C algorithm.

Specifically, at the kth iteration, the proposed procedure first detects an unknown defective sample $s_{\mathrm{DP}}^{(k)}$ that, in general, might have multiple defective bands denoted as $\Theta(s_{\mathrm{DP}}^{(k)})$; then detects the defective bands $\Theta(s_{\mathrm{DP}}^{(k)})$; and finally restores the corrupted data by removing from all the pixels in the line addressed by $s_{\mathrm{DP}}^{(k)}$ the estimates of the detected DPs noise vectors.

Fig. 3 shows that the ACT profile of GRX highlights the defective samples and can therefore be used to detect them. Actually, to avoid the bias due to the background, a median filter is applied to $p_{\mathrm{GRX}}(s)$ in order to decouple the background contribution from the peaks. Specifically, at the kth iteration, $r_{\mathrm{GRX}}^{(k-1)}(s)$ is used instead of $p_{\mathrm{GRX}}^{(k-1)}(s)$ with

$$r_{\mathrm{GRX}}^{(k-1)}(s) = p_{\mathrm{GRX}}^{(k-1)}(s) - \mathrm{median}\left(p_{\mathrm{GRX}}^{(k-1)}(s), 2d+1\right) \quad (11)$$

where $r_{\mathrm{GRX}}^{(k-1)}(s)$ denotes the residual of $p_{\mathrm{GRX}}^{(k-1)}(s)$ after applying the median filter having size $2d + 1$.

The kth defective sample $s_{\mathrm{DP}}^{(k)}$ is obtained by taking the maximum of $r_{\mathrm{GRX}}^{(k-1)}(s)$ and excluding the defective samples $(s_{\mathrm{DP}}^{(1)}, \ldots, s_{\mathrm{DP}}^{(k-1)})$ found in the previous iterations

$$s_{\mathrm{DP}}^{(k)} = \arg\max_{s \in \{s_{\mathrm{DP}}^{(1)}, \ldots, s_{\mathrm{DP}}^{(k-1)}\}} \left\{ r_{\mathrm{GRX}}^{(k-1)}(s) \right\}. \quad (12)$$

Once $s_{\mathrm{DP}}^{(k)}$ has been obtained, a greedy algorithm based on orthogonal matching pursuit (OMP) [24] is adopted to detect the defective bands $\Theta(s_{\mathrm{DP}}^{(k)})$ and to restore the corrupted radiance values in those bands for all the pixels of the lines addressed by $s_{\mathrm{DP}}^{(k)}$.

Recalling (1), (4), and (5), the entire line $\boldsymbol{Y}(s_{\mathrm{DP}}^{(k)}) \in \mathcal{R}^{BxL}$ can be modeled as

$$\boldsymbol{Y}\left(s_{\mathrm{DP}}^{(k)}\right) = \boldsymbol{\Omega} \cdot \mathbf{A}\left(s_{\mathrm{DP}}^{(k)}\right) + \mathbf{E} \cdot \mathbf{D}\left(s_{\mathrm{DP}}^{(k)}\right) + \boldsymbol{N}\left(s_{\mathrm{DP}}^{(k)}\right) \quad (13)$$

where $\mathbf{E}$ is the BxB identity matrix and $\mathbf{A}(s_{\mathrm{DP}}^{(k)})$, $\mathbf{D}(s_{\mathrm{DP}}^{(k)})$, and $\boldsymbol{N}(s_{\mathrm{DP}}^{(k)})$ are the matrices whose columns are $\boldsymbol{a}(s_{\mathrm{DP}}^{(k)}, l)$, $\boldsymbol{d}(s_{\mathrm{DP}}^{(k)}, l)$, and $\boldsymbol{n}(s_{\mathrm{DP}}^{(k)}, l)$ for $l = 1, \ldots, L$, respectively. Notice that $\boldsymbol{d}(s_{\mathrm{DP}}^{(k)}, l)$ for $l = 1, \ldots, L$ are sparse vectors having the nonzero elements in the same few bands $\Theta(s_{\mathrm{DP}}^{(k)})$. Thus, $\mathbf{D}(s_{\mathrm{DP}}^{(k)})$ is also a sparse matrix having few nonzero rows. Both $\Theta(s_{\mathrm{DP}}^{(k)})$ and the values of $\mathbf{D}(s_{\mathrm{DP}}^{(k)})$ in the rows addressed by $\Theta(s_{\mathrm{DP}}^{(k)})$ are unknown. Equation (13) can be rewritten in the noise-whitened space as

$$\boldsymbol{Y}_W\left(s_{\mathrm{DP}}^{(k)}\right) = \boldsymbol{\Omega}_W \cdot \mathbf{A}\left(s_{\mathrm{DP}}^{(k)}\right) + \boldsymbol{E}_W \cdot \mathbf{D}\left(s_{\mathrm{DP}}^{(k)}\right) + \boldsymbol{W}\left(s_{\mathrm{DP}}^{(k)}\right) \quad (14)$$

with

$$\boldsymbol{\Omega}_W = \boldsymbol{\Lambda}^{-\frac{1}{2}} \cdot \boldsymbol{\Omega}, \; \boldsymbol{E}_W = \boldsymbol{\Lambda}^{-\frac{1}{2}} \cdot \mathbf{E} \quad (15)$$

and $\boldsymbol{W}(s_{\mathrm{DP}}^{(k)}) = \boldsymbol{\Lambda}^{-(1/2)} \cdot \boldsymbol{N}(s_{\mathrm{DP}}^{(k)})$ representing the Gaussian noise vector with identity covariance matrix.

Indicating with $R\{\tilde{\Theta}\}$ the cardinality of the generic set of bands $\tilde{\Theta}$, the estimation of $\mathbf{D}(s_{\mathrm{DP}}^{(k)})$ and $\Theta(s_{\mathrm{DP}}^{(k)})$ from $\boldsymbol{Y}_W(s_{\mathrm{DP}}^{(k)})$ can be reformulated as the simultaneous sparse approximation problem [25]

$$\begin{aligned} &\Theta\left(s_{\mathrm{DP}}^{(k)}\right) \\ &\quad = \mathrm{argmin}_{\Theta} R\left\{\tilde{\Theta}\right\} \\ &\qquad \text{s.t.} \left\| \boldsymbol{Y}_W\left(s_{\mathrm{DP}}^{(k)}\right) - \boldsymbol{\Omega}_W \cdot \mathbf{A}\left(s_{\mathrm{DP}}^{(k)}\right) - \boldsymbol{E}_W \cdot \mathbf{D}\left(s_{\mathrm{DP}}^{(k)}\right) \right\|_F < \varepsilon \end{aligned} \quad (16)$$

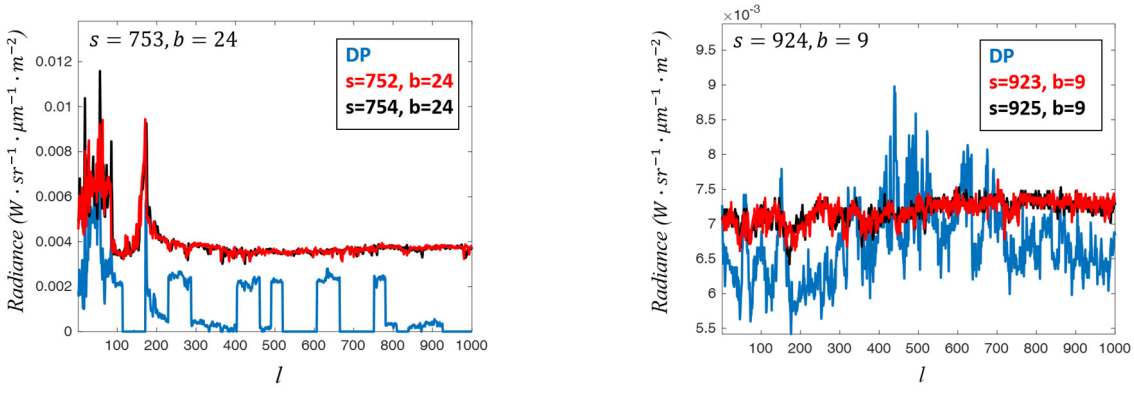
where $||\,||_F$ denotes the Frobenius norm.

The problem in (16) can be attacked by the modified version of the simultaneous-OMP (S-OMP) algorithm (see [25], [26]) described in Section IV-A. The outputs of S-OMP are both $\Theta(s_{\mathrm{DP}}^{(k)})$ and the sparse matrix $\mathbf{D}(s_{\mathrm{DP}}^{(k)})$. The first is used to update $\boldsymbol{\Psi}^{(k)}$, while $\mathbf{D}(s_{\mathrm{DP}}^{(k)})$ is used to restore the corrupted values as

$$\tilde{\boldsymbol{Y}}\left(s_{\mathrm{DP}}^{(k)}\right) = \boldsymbol{Y}\left(s_{\mathrm{DP}}^{(k)}\right) - \mathbf{D}\left(s_{\mathrm{DP}}^{(k)}\right) \quad (17)$$

with $\tilde{\boldsymbol{Y}}(s_{\mathrm{DP}}^{(k)})$ denoting the corrected pixels in the entire ALT direction. Notice that, here again, based on the sparsity of the matrix $\mathbf{D}(s_{\mathrm{DP}}^{(k)})$, (17) modifies only the corrupted bands in $\boldsymbol{Y}(s_{\mathrm{DP}}^{(k)})$.

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(a) (b)

Fig. 4. (a) HRC image values in the line corresponding to $s = 753$ and $b = 24$ and values of the two adjacent lines taken as a reference for the uncorrupted signal. (b) Similar plots for the DP in $s = 924$ and $b = 9$.

The estimates of the signal subspace basis matrix $\boldsymbol{\Omega}$ and the noise covariance matrix $\boldsymbol{\Lambda}$ are obtained by the HySime algorithm applied to all the pixels in the neighborhood Υ of $s_{\mathrm{DP}}^{(k)}$. Υ is defined as in (10), where $\bar{\boldsymbol{\Psi}}(b,s)$ is replaced by $\Psi^{(k-1)}(b,s)$.

$\tilde{\boldsymbol{Y}}(s_{\mathrm{DP}}^{(k)})$ is adopted to update the value of the GRX ACT profile in $s_{\mathrm{DP}}^{(k)}$ as

$$p_{\mathrm{GRX}}^{(k)}\left(s_{\mathrm{DP}}^{(k)}\right) = \frac{1}{L}\left\|\boldsymbol{\Gamma}^{-\frac{1}{2}}\left[\tilde{\boldsymbol{Y}}\left(s_{\mathrm{DP}}^{(k)}\right) - \boldsymbol{\mu}\right]\right\|_F^2. \tag{18}$$

The procedure stops when the maximum of $r_{\mathrm{GRX}}^{(k)}(s)$ with respect to s is below a given threshold α obtained as

$$\alpha = 3 \cdot \mathrm{mad}\left(\boldsymbol{r}_{\mathrm{GRX}}^{(k-1)}\right) \tag{19}$$

where mad() extracts the median absolute deviation of the entire residual at the $(k-1)$th iteration $\boldsymbol{r}_{\mathrm{GRX}}^{(k-1)} = [r_{\mathrm{GRX}}^{(k-1)}(1), \ldots, r_{\mathrm{GRX}}^{(k-1)}(S)]$ and Mad is adopted as an estimator of the standard deviation of the residuals $\boldsymbol{r}_{\mathrm{GRX}}^{(k-1)}$ exploiting its robustness to outliers.

A. Modified S-OMP Algorithm

In order to obtain the solution of the problem in (16), we resort to a slightly modified version of the S-OMP algorithm described in the following, where, for the sake of readability, we omit the argument $s_{\mathrm{DP}}^{(k)}$.

S-OMP is an iterative greedy algorithm that selects at each step the atom of the adopted dictionary that is the most correlated with all the elements of the current residual matrix. The adopted modified version of S-OMP considers as dictionary $\boldsymbol{E}_W = [\boldsymbol{e}_{W,1}, \ldots, \boldsymbol{e}_{W,B}]$ and starts from the residual matrix obtained as

$$\boldsymbol{R}^{(0)} = \boldsymbol{Y}_W - \boldsymbol{\Omega}_W \cdot \boldsymbol{\Omega}_W^{+} \cdot \boldsymbol{Y}_W \tag{20}$$

with $\boldsymbol{\Omega}_W^{+}$ denoting the Moore–Penrose pseudoinverse of $\boldsymbol{\Omega}_W$. Θ is initialized as an empty set ($\Theta^{(0)} = \emptyset$) and $\mathbf{D}$ is initialized as a BxL null matrix [$\mathbf{D}^{(0)} = \mathbf{0}$)].

At the ith iteration, the set of the selected atoms $\Theta^{(i)}$ is updated as $\Theta^{(i)} = \Theta^{(i-1)} \cup t$ with t denoting the atoms of $\boldsymbol{E}_W$ obtained as

$$t = \mathrm{argmax}_{t\in\{1,\ldots,B\}} \frac{\left\|\boldsymbol{e}_{W,\tilde{t}}^T \cdot \boldsymbol{R}^{(i-1)}\right\|_2}{\left\|\boldsymbol{e}_{W,\tilde{t}}\right\|_2} \tag{21}$$

where the symbol ^{T} denotes the vector transpose.

The residual matrix $\boldsymbol{R}^{(i)}$ is updated by

$$\boldsymbol{R}^{(i)} = \boldsymbol{Y}_W - \left[\boldsymbol{\Omega}_W, \boldsymbol{E}_{W,\Theta^{(i)}}\right] \cdot \left[\boldsymbol{\Omega}_W, \boldsymbol{E}_{W,\Theta^{(i)}}\right]^{+} \cdot \boldsymbol{Y}_W \tag{22}$$

with $\boldsymbol{E}_{W,\Theta^{(i)}}$ representing the matrix formed by the atoms $\boldsymbol{e}_{W,t}$ for all $t \in \Theta^{(i)}$.

Notice that

$$\begin{bmatrix} \hat{\boldsymbol{A}}^{(i)} \\ \hat{\boldsymbol{D}}_{\Theta^{(i)}} \end{bmatrix} = \left[\boldsymbol{\Omega}_W, \boldsymbol{E}_{W,\Theta^{(i)}}\right]^{+} \cdot \boldsymbol{Y}_W$$

where $\hat{\boldsymbol{A}}^{(i)}$ is the ML estimate of the matrix $\mathbf{A}$ and $\hat{\boldsymbol{D}}_{\Theta^{(i)}}$ is the ML estimate of the i rows of $\mathbf{D}$ addressed by $\Theta^{(i)}$. Accordingly, the matrix $\mathbf{D}^{(i)}$ is an update by replacing the values in the rows $\Theta^{(i)}$ with $\hat{\boldsymbol{D}}_{\Theta^{(i)}}$.

S-OMP iterates until the stopping condition $\|\boldsymbol{R}^{(i)}\|_F^2 < \varepsilon$ is achieved. The threshold ε is set so as to have a high probability p that the residual matrix $\boldsymbol{R}^{(i)}$ contains only random noise realizations ($\boldsymbol{W}$). In that case, $\|\boldsymbol{R}^{(i)}\|_F^2$ is the sum of squared independent and identically distributed Gaussian random variables with zero mean and unit variance. Therefore, it follows a central chi-square distribution with n degrees of freedom, n being the number of elements of $\boldsymbol{R}^{(i)}$. Accordingly, ε is set as

$$\varepsilon = F_{\chi_n^2}^{-1}(p) \tag{23}$$

where $F_{\chi_n^2}^{-1}()$ is the inverse of the cumulative distribution function of the central chi-square random variable with n degrees of freedom.

It is worth noting that, $\boldsymbol{R}^{(i)} \in \mathcal{R}^{BxL}$, so in theory, $n = BxL$. However, in order to avoid the bias on $\|\boldsymbol{R}^{(i)}\|_F^2$ induced by spectral anomalies not due to DPs having strong components in the orthogonal complement of the useful signal subspace Φ, a censoring mechanism is implemented discarding a given small percentage β of the columns of $\boldsymbol{R}^{(i)}$ with the highest $L2$-norm.

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TABLE I
DPs-DC Algorithm Summary

Algorithm DPs-DC

Input: HRC PRISMA image $\overline{\boldsymbol{y}}(s,l)$ after the application of DPs-C, known DPM $\overline{\boldsymbol{\Psi}}$, dimension of the median filter $2d+1$, dimension of the local neighborhood for signal subspace estimation $2m+1$

Output: DPs corrected image $\widetilde{\boldsymbol{y}}(s,l)$, estimated DPM $\boldsymbol{\Psi}$

Initialize: $p_{GRX}^{(0)}(s)$ ACT profile of the GRX applied to the data $\overline{\boldsymbol{y}}(s,l)$, $\boldsymbol{\Psi}^{(0)}=\overline{\boldsymbol{\Psi}}$

***k*th iteration**:

Step 1: Compute the residual $r_{GRX}^{(k-1)}(s)$ by eq. (11) and the threshold α by eq. (19)

Step 2: Detect defective sample $s_{DP}^{(k)}$ by eq. (12)
Estimate the local basis matrix $\boldsymbol{\Omega}$ and the noise covariance matrix $\boldsymbol{\Lambda}$ by the HySime algorithm applied to the neighborhood Υ of $s_{DP}^{(k)}$

Step 3: Detect the defective bands $\Theta\left(s_{DP}^{(k)}\right)$ and the DPs noise $\mathbf{D}\left(s_{DP}^{(k)}\right)$ by the modified version of S-OMP

Step 4: Restore the corrupted values through eq. (17) and update the DPM $\boldsymbol{\Psi}^{(k)}$

Step 5: Update the ACT profile of GRX in the defective sample $s_{DP}^{(k)}$ through eq. (18)

Stopping rule:

repeat steps 1-5 until $\max_{s\notin\left\{s_{DP}^{(1)}\ldots,s_{DP}^{(k)}\right\}}\left\{r_{GRX}^{(k)}(s)\right\}>\alpha$

TABLE II
Modified S-OMP Algorithm Summary

Algorithm Modified S-OMP

Input: $\boldsymbol{Y}_W, \boldsymbol{\Omega}_W, \boldsymbol{E}_W, p, \beta$

Output: $\Theta, \boldsymbol{D}$

Initialize: $\boldsymbol{R}^{(0)}=\boldsymbol{Y}_W-\boldsymbol{\Omega}_W\cdot\boldsymbol{\Omega}_W^{+}\cdot\boldsymbol{Y}_W$, $\Theta^{(0)}=\emptyset$, $\boldsymbol{D}^{(0)}=\mathbf{0}$

***i*th iteration**:

Step 1: select the atom t of the dictionary $\boldsymbol{E}_W$ by eq. (21) and update $\Theta^{(i)}$ as $\Theta^{(i)}=\Theta^{(i-1)}\cup t$

Step 2: update the residual matrix by eq. (22)
$$\boldsymbol{R}^{(i)}=\boldsymbol{Y}_W-\left[\boldsymbol{\Omega}_W,\boldsymbol{E}_{W,\Theta^{(i)}}\right]\cdot\left[\boldsymbol{\Omega}_W,\boldsymbol{E}_{W,\Theta^{(i)}}\right]^{+}\cdot\boldsymbol{Y}_W$$

Step 3: update the matrix $\boldsymbol{D}^{(i)}$ by replacing the row addressed by $\Theta^{(i)}$ with $\widehat{\mathbf{D}}_{\Theta^{(i)}}$

Step 4: calculate $\left\|\boldsymbol{R}^{(i)}\right\|_F^2$ by discarding the βL columns of $\boldsymbol{R}^{(i)}$ with the highest L2 norm (censoring) and compute the number n of the retained columns

Stopping rule:

repeat steps 1-4 until $\left\|\boldsymbol{R}^{(i)}\right\|_F^2>F_{\chi_n^2}^{-1}(p)$

B. DPs-DC Algorithm Summary

The summary of the DPs-DC algorithm is shown in Table I. Furthermore, we report the summary of the modified S-OMP procedure that is an integral part of DPs-DC in Table II.

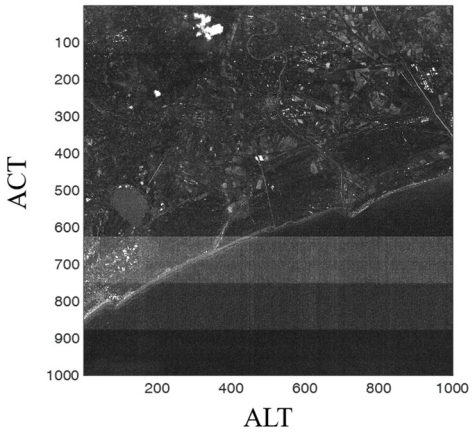


Fig. 5. Output of the GRX obtained on the VNIR HRC image of PISA after the application of DPs-DC. It should be compared with Fig. 2(b) to appreciate the improvement.

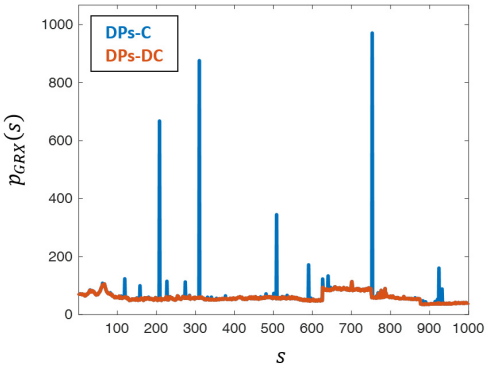


Fig. 6. ACT profiles of the GRX output obtained on the VNIR HRC image after the correction of the known DPs (DPs-C algorithm) and after the application of DPs-DC.

The proposed algorithm has four free parameters denoted as m, d, p, and β. m determines the size of the neighborhood adopted for local estimation of the signal subspace basis matrix $\boldsymbol{\Omega}$ via HySime. It should be chosen as a compromise between two conflicting requirements. On the one hand, it should be small enough to obtain a compact description for the signal subspace by exploiting the strong local spatial correlation that is generally experienced in hyperspectral images. On the other hand, it should be large enough to avoid the bias introduced on the estimate of $\boldsymbol{\Omega}$ by the potential DPs spatially close to the one being analyzed and not already detected. In all the experiments performed on PRISMA data, we found that $m=3$ can be considered a good choice. The parameter d determines the length of the median filter used to decouple the background from the peaks in the ACT profile of the GRX output. It should be set according to the maximum expected number of adjacent defective samples in the analyzed image. In the considered PRISMA data, we found that $d=3$ provides satisfactory results. The parameter p is adopted by the stopping rule of the modified S-OMP algorithm and represents the probability that the residual matrix $\boldsymbol{R}^{(i)}$ is made by Gaussian noise realizations. In all the experiments presented in this article, it is set to 0.9. The parameter β is used by the censoring mechanism used in the stopping rule of the modified S-OMP algorithm. Censoring is necessary to prevent small spectral anomalies in the scene

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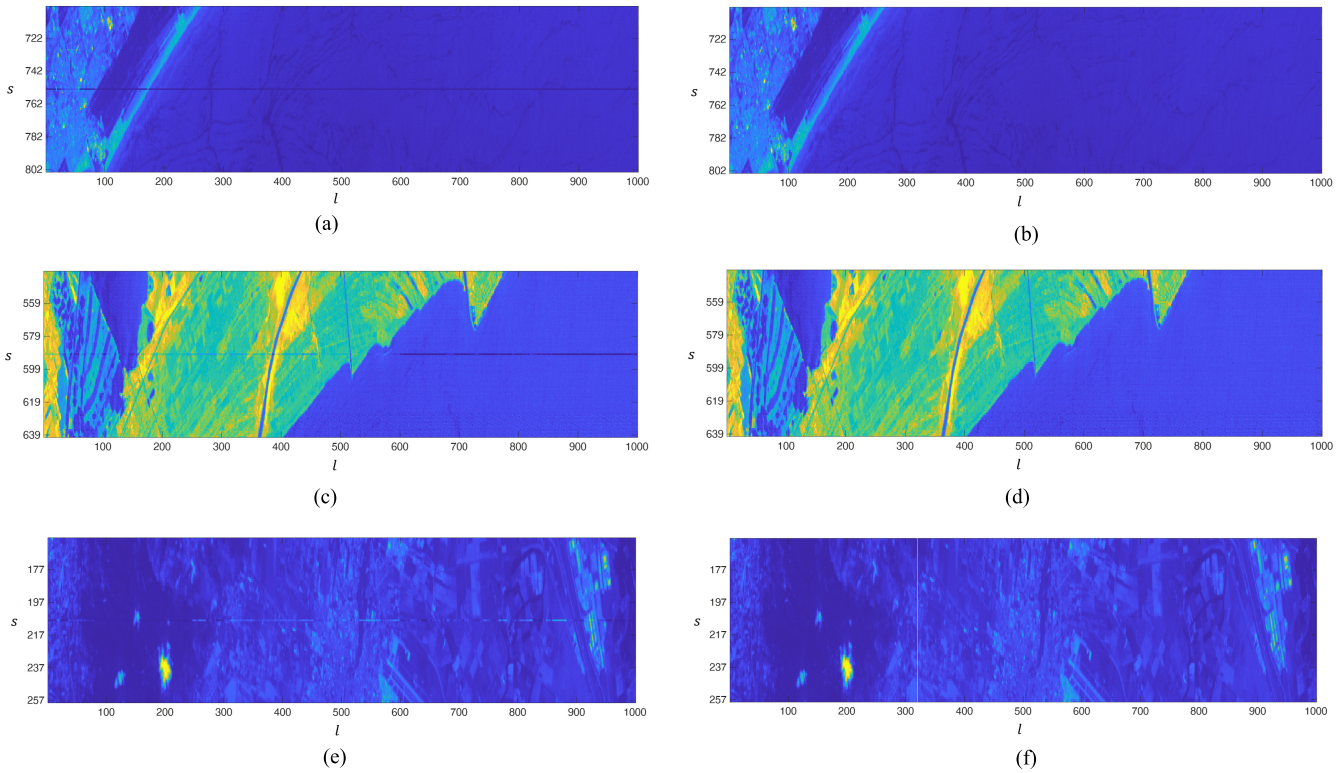


Fig. 7. (a) Original VNIR HRC image, DP in $s = 753$ and $b = 24$; (b) image after DPs-DC, in $s = 753$ and $b = 24$; (c) original VNIR HRC image, DP in $s = 590$ and $b = 61$; (d) image after DPs-DC, in $s = 590$ and $b = 61$; (e) original VNIR HRC image, DP in $s = 208$ and $b = 13$; and (f) image after DPs-DC, in $s = 208$ and $b = 13$.

from being mistaken for DPs. In all the experiments presented in this article, β is set to 10%.

We conclude this section by showing the results of the DPs-DC algorithm on the VNIR PRISMA HRC image considered in Sections I and III. Fig. 5 shows the GRX output of the image provided by the proposed algorithm. Comparing Fig. 5 with Fig. 2(b), it is evident the effectiveness of the proposed algorithm. All the bright stripes present in Fig. 2(b) and due to unknown DPs vanish when we consider the GRX output of the image provided by DPs-DC. This is also confirmed by the comparison, as shown in Fig. 6, between the ACT profiles of the GRX outputs obtained before (blue line) and after (red line) the application of DPs-DC. In Fig. 6, the improvement made by the DPs-DC algorithm on the analyzed data is quite evident.

It is worth noting that the peaks corresponding to the two known defective samples in $s = 753$ and $s = 924$ that were analyzed in Section III are not present in the ACT profile of the GRX applied to the image provided by the DPs-DC algorithm. This shows that the proposed algorithm correctly restores the corrupted radiance values corresponding to the additional DPs in $s = 753$ and $b = 24$ and $s = 924$ and $b = 9$ (not reported in $\bar{\mathbf{\Psi}}$).

As an example, with reference to the DP detected by the proposed procedure at $s = 753$ and $b = 24$, Fig. 7(a) and (b) shows the band $b = 24$ of the original HRC data and the image after DP correction, respectively. In both the figures, the neighborhood centered on the bad sample $s = 753$ is represented. In Fig. 7(a), we can clearly appreciate the presence of the stripe due to the DP. It completely vanishes in the image obtained by the proposed procedure [Fig. 7(b)]. Similar examples are reported in Fig. 7(c) and (d) with reference to the DP at $s = 590$ and $b = 61$ and in Fig. 7(e) and (f) with reference to the DP at $s = 208$ and $b = 13$.

For the analyzed image, the number of known DPs is 61 corresponding to 0.0968% of the total number of detectors ($BxS = 63 \times 10^3$) in the acquisition matrix. The number of additional DPs detected by the proposed algorithm is 51. Thus, in the image, the DPs are 0.178% of the total pixels.

V. Experimental Results

In this section, we present and discuss the results obtained by applying both the DPs-C and DPs-DC algorithms to different PRISMA images. Specifically, we first complete the analysis of the PRISMA data introduced in Sections I, III and IV by presenting the results obtained on the image acquired by the SWIR spectrometer. Then, we define two performance metrics and introduce a set of PRISMA images for the numerical performance analysis of the proposed algorithms. Finally, we present and discuss the results obtained on the selected dataset.

A. Analysis of the Results Obtained on the Image Acquired by the SWIR Spectrometer

In this section, with reference to the PRISMA data acquired in the region around the town of PISA on June 10, 2021, we analyze the results provided by the proposed algorithms on the SWIR HRC hyperspectral image. Specifically, in Fig. 8, we show the GRX outputs obtained on both the original HRC

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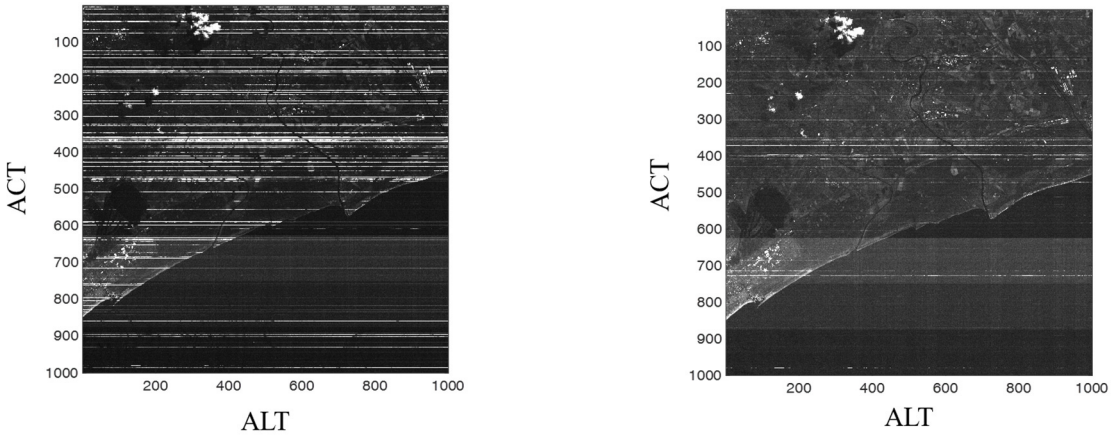


Fig. 8. Output of the GRX obtained on (a) original SWIR HRC image of PISA and (b) SWIR HRC image after DPs-C application.

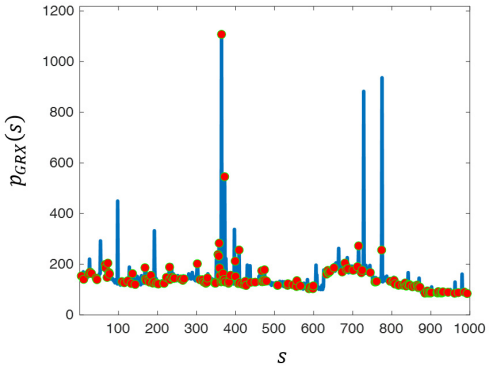


Fig. 9. ACT profile of the GRX $[p_{\mathrm{GRX}}(s)]$ output obtained on the SWIR HRC image after the application of DPs-C. Red plots indicate the defective samples reported in $\bar{\Psi}$.

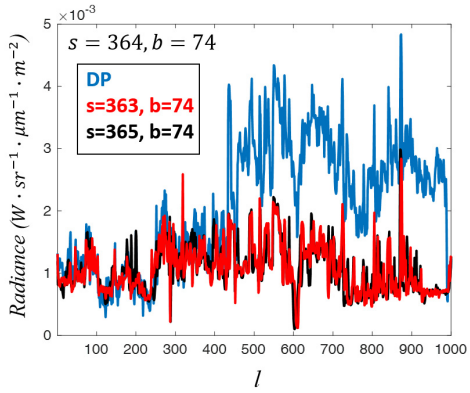


Fig. 10. HRC image values in the line corresponding to $s = 364$ and $b = 74$ and values of the two adjacent lines taken as a reference for the uncorrupted signal.

image [Fig. 8(a)] and that provided by the DPs-C algorithm where the DPs indicated by the kDPM ($\bar{\Psi}$) are corrected [Fig. 8(b)].

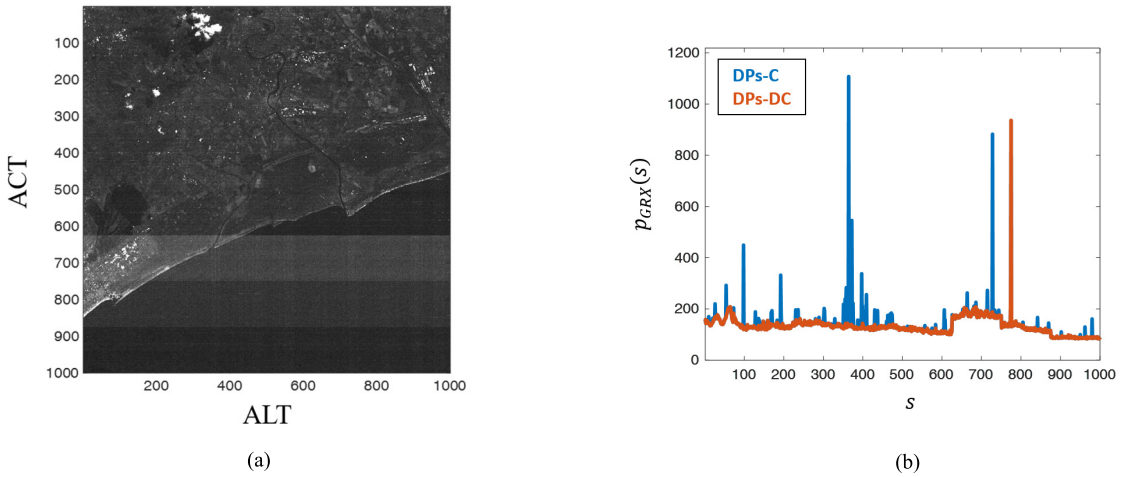
As in the VNIR image, we see that most of the bright stripes in Fig. 8(a) vanish after the application of the DPs-C algorithm. This is also confirmed by the results in Fig. 9, which shows the ACT profile of the GRX after the application of DPs-C algorithm and highlights the position of known defective samples reported in $\bar{\Psi}$.

Here, again, some of the peaks corresponding to known defective samples are not removed by the DPs-C algorithm because, as discussed in Section III, some defective samples in $\bar{\Psi}$ have more defective bands than those indicated by $\bar{\Psi}$. As an example, let us consider the defective sample at $s = 364$, which is the maximum in Fig. 9. For $s = 364$, $\bar{\Psi}$ reports two defective bands at $b = 72$ and $b = 73$. We found an additional DP at $b = 74$. In Fig. 10, we plot the entire line of the HRC image corresponding to $s = 364$ and $b = 74$ together with the two adjacent lines ($s = 363$ and $s = 365$). The anomalous signal behavior from line 450 to line 1000 is quite evident and, according to the classes defined in Section II, this DP is an example of irregular DPs.

Fig. 11(a) and (b) shows the results obtained by applying the DPs-DC algorithm. Specifically, they show the GRX output [Fig. 11(a)] and its ACT profile [Fig. 11(b)] obtained on the image provided by the proposed algorithm. For completeness, Fig. 11(b) also shows the GRX profile obtained after the application of DPs-C. Comparing Fig. 8(b) with Fig. 11(a), we can appreciate the improvement due to the application of DPs-DC algorithm. This is further confirmed by comparing the GRX profiles in Fig. 11(b), where almost all the peaks in the profile obtained on the image processed by the DPs-C algorithm disappear after the application of DPs-DC. Only a peak remains at $s = 775$. DPs-DC has not detected any DPs with ACT coordinate $s = 775$. This can be explained by observing that the peak at $s = 775$ is not due to a DP but rather to a small size spectral anomaly that is correctly ignored by DPs-DC due to the censoring introduced in the modified S-OMP procedure. To convince the reader, in Fig. 12, we plot the entire line at $s = 775$ of the GRX output obtained on the image after the application of DPs-DC that has exactly the same values as that of GRX applied to the original HRC image.

In Fig. 12, we use the logarithmic scale for the y-axis and highlight (red dashed line) the 99th percentile of the entire GRX output. We see that only few pixels are above the 99th percentile, those in the lines around 13 and around 120. These are clearly small spectral anomalies whose high anomaly

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(a) (b)

Fig. 11. SWIR HRC image of PISA after the application of the DPs-DC algorithm: (a) output of the GRX and (b) ACT profile of the GRX output.

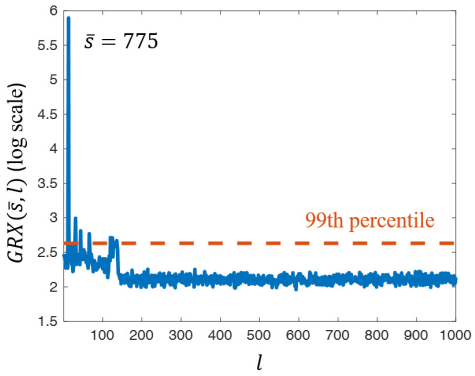


Fig. 12. GRX values (in logarithmic scale) of the entire line at $s = 775$ obtained on the image provided by the DPs-DC algorithm. The red dashed line indicates the 99th percentile of the entire GRX output.

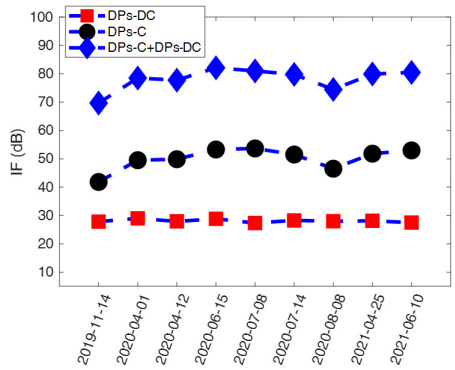


Fig. 13. IF values in decibels obtained on the considered VNIR PRISMA images.

degree biases the GRX profile producing the peak at $s = 775$ both in the profile obtained on the original HRC image and in the image provided by the DPs-DC algorithm.

B. Performance Metrics and Considered PRISMA Data

In order to quantitatively evaluate the performance of both DPs-C and DPs-DC algorithms, we use two performance metrics.

The first metric, called improvement factor (IF) [27], [28], measures the improvement due to the application of a certain algorithm in terms of removal of the persistent anomalies in the ALT direction. IF is defined as

$$\mathrm{IF} = 10\log_{10}\frac{\sum_{s=1}^{S}\left[p_{\mathrm{GRX}}^{(\mathrm{IN})}(s) - p_{\mathrm{GRX}}^{(\mathrm{BKG})}(s)\right]^2}{\sum_{s=1}^{S}\left[p_{\mathrm{GRX}}^{(\mathrm{OUT})}(s) - p_{\mathrm{GRX}}^{(\mathrm{BKG})}(s)\right]^2} \tag{24}$$

where $p_{\mathrm{GRX}}^{(\mathrm{IN})}(s)$ and $p_{\mathrm{GRX}}^{(\mathrm{OUT})}(s)$ are the ACT profiles of the GRX outputs obtained on the image before and after the application of the DPs-C algorithm, respectively. $p_{\mathrm{GRX}}^{(\mathrm{BKG})}(s)$ is the estimated GRX profile of the image not affected by noise due to the DPs and is obtained by filtering out from $p_{\mathrm{GRX}}^{(\mathrm{OUT})}(s)$ the peaks due to the DPs by means of a median filter of length $2d + 1$. In all the considered experiments, we set $d = 3$.

Positive values of IF indicate an improvement in the image quality provided by the algorithm. The higher the IF value, the greater the improvement produced by the algorithm.

The radiometric quality of the estimated radiance in the DPs cannot be accurately assessed on real data because true values are not available. However, a fairly reliable assessment of the reconstruction accuracy can be obtained using the following metric based on the nearest neighbor paradigm.

Let us focus on the DP with coordinates $(b_{\mathrm{DP}}, s_{\mathrm{DP}})$ and denote as $s_1 < s_{\mathrm{DP}}$ and $s_2 > s_{\mathrm{DP}}$ the closest samples to s_{DP} not detected as defective and as $b_1 < b_{\mathrm{DP}}$ and $b_2 > b_{\mathrm{DP}}$ the closest bands to b_{DP} not detected as defective.

For each line l, we consider the quantity $\vartheta_{b_{\mathrm{DP}}}(s_{\mathrm{DP}}, l)$ defined as the minimum of the relative squared errors among the radiance value $\hat{y}_{b_{\mathrm{DP}}}(s_{\mathrm{DP}}, l)$ estimated by the algorithm and the radiance values of the image pixels addressed by the sample and band coordinates previously introduced, namely

$$\vartheta_{b_{\mathrm{DP}}}(s_{\mathrm{DP}}, l) = \left[\frac{\hat{y}_{b_{\mathrm{DP}}}(s_{\mathrm{DP}}, l) - y_b(s, l)}{y_b(s, l)}\right]^2 \quad \text{with}$$
$$I \equiv \{(b_1, s_1), (b_1, s_2), (b_2, s_1), (b_2, s_2)\}. \tag{25}$$

The values of $\vartheta_{b_{\mathrm{DP}}}(s_{\mathrm{DP}}, l)$ are calculated for each line l and their average value named root relative mean squared error (RRMSE) is adopted as a synthetic measure of the algorithm's

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TABLE III

LIST OF THE PRISMA DATA ADOPTED IN THE PRESENTED EXPERIMENTS. FOR EACH DATA, THE "HDF5" FILE NAME AND THE ACQUISITION DATE ARE REPORTED

File name	Acquisition date year-month-day
PRS_L1_STD_OFFL_20191114102719_20191114102723_0001.he5	2019-11-14
PRS_L1_STD_OFFL_20200401102434_20200401102438_0001.he5	2020-04-01
PRS_L1_STD_OFFL_20200412101431_20200412101435_0001.he5	2020-04-12
PRS_L1_STD_OFFL_20200615101742_20200615101746_0001.he5	2020-06-15
PRS_L1_STD_OFFL_20200708101415_20200708101419_0001.he5	2020-07-08
PRS_L1_STD_OFFL_20200714101730_20200714101734_0001.he5	2020-07-14
PRS_L1_STD_OFFL_20200808104714_20200808104718_0001.he5	2020-08-08
PRS_L1_STD_OFFL_20210425102709_20210425102713_0001.he5	2021-04-25
PRS_L1_STD_OFFL_20210610102029_20210610102033_0001.he5	2021-06-10

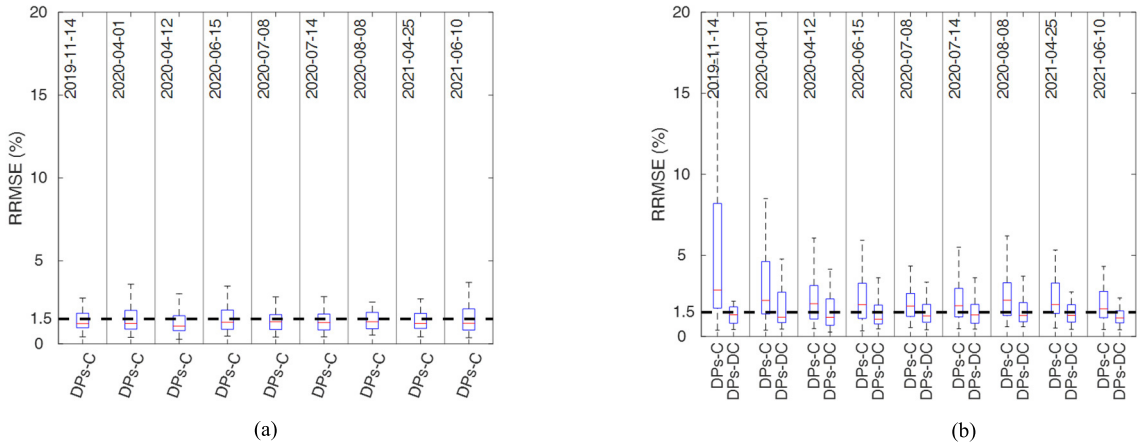


Fig. 14. (a) RRMSE for the known DPs after the application of the DPs-C algorithm to the VNIR PRISMA images. (b) RRMSE for the detected DPs before and after the application of the DPs-DC algorithm to the VNIR PRISMA images.

performance for a given DP

$$\mathrm{RRMSE}(s_{\mathrm{DP}}, b_{\mathrm{DP}}) = \sqrt{\frac{1}{L}\sum_{l=1}^{L} \vartheta_{b_{\mathrm{DP}}}(s_{\mathrm{DP}}, l)}. \qquad (26)$$

Low values of the RRMSE denote high quality of the restored radiance values.

The two metrics IF and RRMSE are used to evaluate the performance of the DPs-C and DPs-DC algorithms on nine PRISMA hyperspectral images acquired in 2019, 2020, and 2021. The names of the "hdf5" files and their acquisition dates are listed in Table III. The last image in Table III is the one collected over the town of Pisa used to illustrate the results in the previous sections.

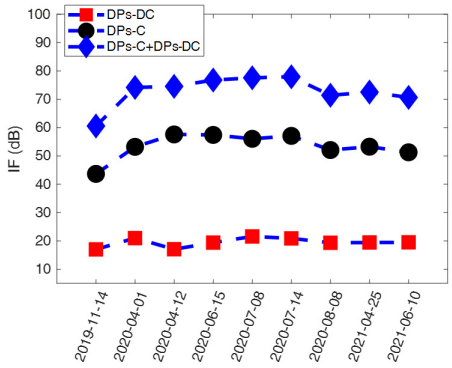


Fig. 15. IF values in decibels obtained on the considered SWIR PRISMA images.

C. Numerical Performance Analysis

For each PRISMA image in Table III, both the DPs-C and DPs-DC algorithms are applied to the two images acquired by the VNIR and the SWIR spectrometers. Of course, DPs-C is applied to the original HRC images and DPs-DC is applied to the outputs provided by DPs-C. Therefore, to compute IF values for DPs-C, $p^{(\mathrm{IN})}_{\mathrm{GRX}}(s)$ is evaluated on the original HRC image. Instead, for DPs-DC, $p^{(\mathrm{IN})}_{\mathrm{GRX}}(s)$ is evaluated on the output of DPs-C.

As to RRMSE, in the case of DPs-C, it is calculated for each image by considering all the DPs in the kDPM. Conversely, in the case of DPs-DC, RRMSE is evaluated by considering all the DPs detected by the procedure and not reported in the kDPM. In this case, RRMSE values are also evaluated for the input image in order to show the improvement due to the application of the DPs-DC algorithm.

In Fig. 13, we show the IF values for the VNIR spectrometer provided by both DPs-C and DPs-DC. We also show the IF values of the entire procedure consisting of the sequential application of DPs-C and DPs-DC. They are calculated by considering $p^{(\mathrm{IN})}_{\mathrm{GRX}}(s)$ obtained on the original HRC

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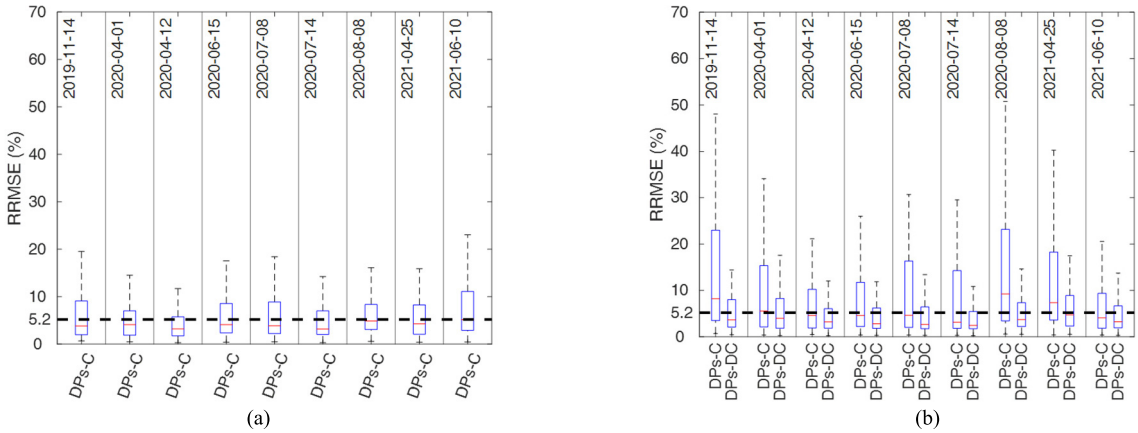


Fig. 16. (a) RRMSE for the known DPs after the application of the DPs-C algorithm to the SWIR PRISMA images. (b) RRMSE for the detected DPs before and after the application of the DPs-DC algorithm to the SWIR PRISMA images.

image and $p_{\mathrm{GRX}}^{(\mathrm{OUT})}(s)$ evaluated on the output of the DPs-DC algorithm.

The high values of IF obtained on all the images confirm the improvement of the image quality achieved by the proposed procedures. Specifically, IF values achieved by DPs-C are higher than 40 dB, while those obtained by DPs-DC are of the order of 28 dB. The overall IF values obtained by the sequential application of the two algorithms, for all the considered image, are above 70 dB.

The higher IF values provided by DPs-C compared to DPs-DC can be explained by considering that the peaks corresponding to the known DPs generally have values higher than those of the unknown DPs. This is due to the fact that the radiance values in the known DPs are forced to zero by the L1 PRISMA processor, so their degrees of anomaly measured by GRX are higher than those of the other DPs.

In Fig. 14, we show the boxplot of the RRMSE (in %) for all the DPs in each of the considered images. Each box plot provides five indexes related to the distribution of the RRMSE values: the red line is the median, the edges of the box are the 25th percentile and the 75th percentile, and the whiskers are the minimum and the maximum values.

In Fig. 14(a), we show the results of the DPs-C algorithms, while in Fig. 14(b), we show the results obtained by applying DPs-DC. In the latter case, we compare RRMSE values obtained before and after the application of the algorithms. Of course, we expect the RRMSE before applying the correction algorithm to be higher due to the anomalous behavior of the defective detectors. Notice that the same comparison makes no sense in the case of DPs-C algorithm. In fact, since the L1 PRISMA processor forces the radiance values in the known DPs to 0, the RRMSE before applying the algorithm has a unitary value (100%) for all of them.

Results in Fig. 14 demonstrate the effectiveness of the proposed algorithms in restoring the missing/corrupted radiance values in both known and detected DPs. In all the considered images, both the algorithms provide RRMSE values much lower than 5% for all the DPs. The median RRMSE values per image are lower than 1.5%. Focusing on Fig. 14(b), we can clearly appreciate the improvement in terms of RRMSE values between input and output images.

Similar results are obtained by applying DPs-C and DPs-DC to the SWIR HRC images. Fig. 15 shows the IF values provided by DPs-C, DPs-DC, and the sequential applications of the two algorithms. IF values achieved by DPs-C are higher than 40 dB, while those obtained by DPs-DC are higher than 17 dB. The overall IF values obtained by the sequential application of the two algorithms, for all the images considered, are higher than 60 dB.

As to the RRMSE, Fig. 16(a) shows the values obtained for the known DPs in all the considered images. For almost all the DPs, RRMSE is lower than 10% and the median RRMSE is below 5.2% in all the images. Fig. 16(b) compares the RRMSE in the DPs detected by DPs-DC before and after the application of the corrupted values restoration procedure. The RRMSE values after the application of the algorithm are in the same order of magnitude of those obtained for the known DPs. Furthermore, Fig. 16(b) highlights the significant improvement introduced by the proposed algorithm.

It is worth noting that the higher RRMSE values obtained in the SWIR images compared to the VNIR ones are probably due to the lower signal-to-noise ratio that characterizes the PRISMA bands within the SWIR spectral range [14].

As to the percentage of DPs in the considered PRISMA images, in Fig. 17, we show separately for the VNIR [Fig. 17(a)] and SWIR images [Fig. 17(b)], the percentages of DPs in the kDPM, and those of the DPs detected by the DPs-DC algorithm. For completeness, we also show the overall percentage of DPs in each image. Fig. 17 shows that the percentage of DPs detected in the VNIR and SWIR images is approximately the same. In total, in all the images considered, the number of DPs is less than 0.3% of the elements of acquisition matrix for both VNIR and SWIR spectrometers.

We conclude this section by providing the computational time for the complete procedure based on the sequential application of DPs-C and DPs-DC. Specifically, using a MATLAB implementation of the procedure on a personal computer equipped with Intel Core i9 CPU at 2.5 GHz and with 64 GB

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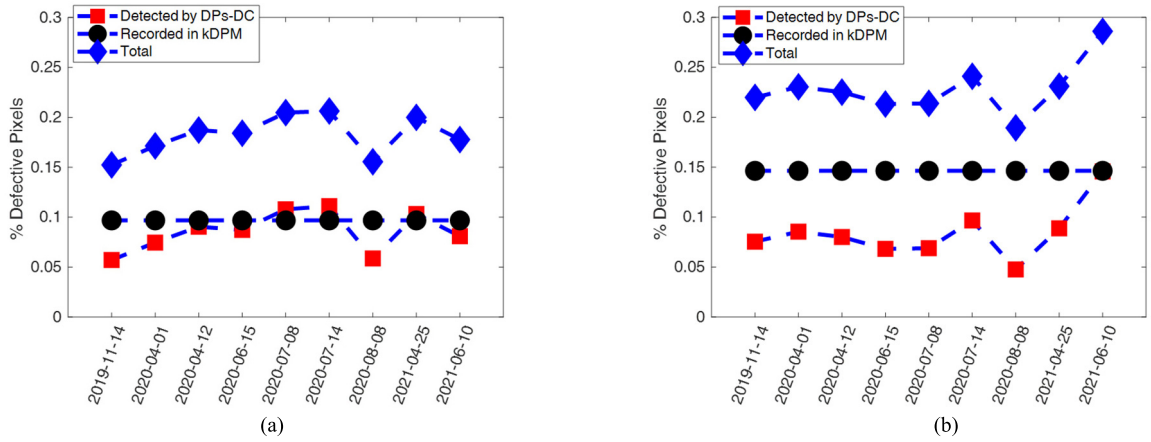


Fig. 17. Percentage of known DPs, detected DPs, and total for (a) VNIR PRISMA images and (b) SWIR PRISMA images.

of RAM, we found an average computing time of 4 s for VNIR images and 18 s for SWIR images.

VI. Conclusion

In this article, we have presented a new procedure to significantly improve the quality of satellite hyperspectral images by reducing the deleterious effects of the nonuniformity noise generated by defective detectors in the acquisition matrix. The procedure is applied to the L1 HRC PRISMA product to avoid that defective radiance values in the faulty detectors are being spread by the spatial/spectral resampling required to compensate smile and keystone and to get the L1 HCO product.

The proposed technique allows for the detection of residual DPs not reported by the in-flight calibration process. For this purpose, both the high sensitivity of GRX to anomalous spectra and the persistence of the anomalies generated by the DPs in the ALT direction are exploited. GRX becomes the guide for unsupervised bad sample detection, while the bad bands are detected by using a modified version of the S-OMP algorithm.

Only the missing values in the DPs are replaced with estimates obtained by regression on an *ad hoc* model of the hyperspectral image. Estimation of the missing values is performed with a new method which, being based on a realistic subspace model, reduces the dimensionality of the problem, and makes the estimates less prone to the noise sources and numerical problems. This allows us to properly fix the missing values avoiding the common assumption that DPs removal can be accomplished by estimating constant gains and offsets.

It is worth noting that the proposed approach differs from most of the existing methods on this subject. In fact, due to the preliminary detection of the residual DPs, we only repair the missing values leaving untouched the rest of the values recorded in the detection matrix. Furthermore, as far as we know, the idea of exploiting the GRX statistic to guide the detection of DPs and, in particular, to detect defective samples has never been proposed to date.

The performance of the proposed procedure has been investigated on a set of nine PRISMA images spanning an interval of about two years. The effectiveness has been demonstrated by showing the improvement in the GRX map. Specifically, the effective removal of the stripes due to DPs, exemplified in Figs. 5 and 11 and also confirmed in all the images of the dataset, is by itself a demonstration of the strength of the proposed method. However, to quantify the improvement, both in terms of ability of removing the stripes generated by DPs and in terms of radiometric accuracy of the estimates of the missing values, we have defined two indexes called IF and RRMSE, respectively. The analysis of the two indexes confirmed the good performance of the method.

The proposed procedure is fully unsupervised. In fact, though it needs the setting of four parameters (m, d, p, and β), their values have been chosen once and for all on the basis of solid physical and statistical assumptions. Remarkably, the same parameter values have yielded good results in all the datasets, thus avoiding the need of tuning them to the specific image.

The analysis of GRX output allowed us to identify a sort of banding effect, which constantly occurs in all the analyzed images. Of course, such nonuniformity source cannot be ascribed to defects of single detectors and it is generated by different causes than those investigated in this work. We thus leave the analysis and the mitigation of such banding effect as a topic for future work.

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